

CHENGDU WENJIANG, China

WMO: 561870

Lat: 30.7489N

Lon: 103.8611E

Elev: 1821

StdP: 13.75

Time Zone: 8.00 (E08)

Period: 05-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 30.9	(c) 33.1	(d) 21.9	(e) 17.6	(f) 42.4	(g) 25.4	(h) 20.8	(i) 40.6	(j) 9.6	(k) 43.7	(l) 8.7	(m) 45.7	(n) 1.7	(o) 340	(p) 0.303

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 13.6	(c) 91.7	(d) 76.0	(e) 89.6	(f) 74.8	(g) 87.5	(h) 73.9	(i) 79.0	(j) 86.6	(k) 77.9	(l) 84.9	(m) 76.9	(n) 83.3	(o) 3.7	(p) 160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
77.2	151.4	82.1	76.2	146.4	81.2	75.2	141.7	80.4	44.3	86.9	43.1	84.9	42.1	83.5	84.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 9.3	(b) 7.7	(c) 6.9	(e) 26.4	(f) 95.6	(g) 2.5	(h) 1.6	(i) 24.6	(j) 96.8	(k) 23.1	(l) 97.7	(m) 21.8	(n) 98.7	(o) 19.9	(p) 99.8		
			DB													
			WB													
				25.6	80.9	2.5	1.7	23.8	82.0	22.3	83.0	20.9	83.9	19.1	85.1	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	62.5	42.4	46.7	55.2	64.4	71.0	75.7	78.8	79.1	71.9	64.1	54.9	45.2
	DBStd	13.55	4.19	5.20	6.06	5.83	4.37	3.70	3.47	4.03	4.54	4.44	5.17	3.95
	HDD50	544	236	118	20	0	0	0	0	0	0	0	16	154
	HDD65	2596	699	513	308	81	6	0	0	0	3	70	302	614
	CDD50	5117	1	25	180	432	649	770	894	901	658	437	164	6
	CDD65	1694	0	0	3	62	191	320	429	436	211	42	0	0
	CDH74	11956	0	0	29	416	1121	2142	3525	3656	950	117	0	0
	CDH80	3824	0	0	2	79	279	622	1219	1382	228	13	0	0
(14) Wind	WSAvg	3.0	2.9	3.1	3.3	3.4	3.5	3.2	3.2	3.1	3.0	2.8	2.6	2.5
(15) Precipitation	PrecAvg	34.9	0.4	0.5	0.9	2.0	2.9	4.6	8.5	8.1	4.6	1.6	0.6	0.3
	PrecMax	53.7	0.8	1.2	2.0	5.2	6.8	9.2	21.7	14.9	9.6	3.0	1.5	0.7
	PrecMin	25.2	0.0	0.0	0.1	0.4	0.8	1.1	3.0	3.3	1.3	0.6	0.1	0.0
	PrecStd	6.5	0.2	0.4	0.5	0.9	1.3	2.1	4.1	3.0	2.4	0.7	0.3	0.2
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	60.5	67.0	77.3	86.2	89.4	90.9	93.8	94.9	89.3	81.7	71.4	59.6
		MCWB	47.6	52.7	60.3	64.1	68.2	71.7	76.6	76.4	74.4	67.6	59.7	48.9
	2%	DB	55.5	62.7	73.0	82.1	86.1	88.3	91.1	92.5	85.6	77.2	67.9	56.4
		MCWB	45.7	51.5	58.0	64.3	67.1	71.5	76.4	76.9	71.7	65.3	58.5	47.8
	5%	DB	52.1	58.8	69.3	78.9	83.1	86.0	89.1	90.1	82.5	73.7	65.1	54.0
		MCWB	44.8	49.7	56.8	63.0	66.5	71.4	75.8	76.3	70.8	64.0	57.3	47.0
	10%	DB	49.6	55.7	65.8	75.5	80.3	83.8	86.7	87.6	79.3	70.6	62.5	52.0
		MCWB	43.8	48.4	55.6	62.3	66.2	71.4	75.7	75.5	70.0	63.7	56.3	46.4
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	49.3	54.1	63.5	69.6	73.4	78.3	80.7	80.2	77.5	70.8	61.2	52.0
		MCDB	56.2	63.4	72.5	79.7	81.1	83.5	88.8	89.1	84.5	76.0	67.8	56.0
	2%	WB	47.6	52.4	60.0	66.6	71.1	76.2	79.2	79.0	75.2	68.2	60.0	50.1
		MCDB	53.2	60.2	68.6	76.0	78.9	81.4	86.5	87.2	81.0	72.6	65.5	53.5
	5%	WB	46.0	50.8	58.3	65.1	69.8	74.7	78.1	78.0	73.4	66.4	58.8	48.8
		MCDB	50.2	57.5	66.0	74.2	77.6	80.7	84.8	85.6	78.7	71.0	63.1	52.1
	10%	WB	44.7	49.1	56.7	63.5	68.5	73.4	77.1	77.1	71.6	64.9	57.6	47.4
		MCDB	48.6	54.9	64.4	72.1	76.2	79.8	83.4	83.9	76.9	69.4	61.5	50.7
(35) Mean Daily Temperature Range	5% DB	MDBR	11.4	12.0	14.3	15.8	14.9	12.9	13.0	13.6	10.4	9.7	10.4	11.6
		MCDBR	17.5	19.3	21.6	22.6	21.0	18.3	18.3	18.2	15.8	15.6	15.1	16.0
		MCWBR	9.5	9.5	8.3	7.1	6.4	5.8	6.4	5.8	4.5	5.4	6.9	8.8
	5% WB	MCDBR	15.6	17.1	17.6	18.4	15.7	13.5	14.9	15.6	12.2	12.0	12.3	12.5
		MCWBR	9.0	8.8	7.3	6.4	6.3	5.2	6.0	5.9	4.9	5.3	6.1	7.4
(40) Clear-Sky Solar Irradiance	taub		0.663	0.709	0.743	0.733	0.696	0.663	0.691	0.720	0.732	0.735	0.714	0.670
	taud		1.496	1.450	1.431	1.490	1.599	1.689	1.650	1.583	1.538	1.499	1.485	1.517
	Ebn at Noon		171	180	189	200	209	215	209	200	189	173	158	161
	Edh at Noon		78	89	97	94	85	77	80	85	86	83	78	73
(44) All-Sky Solar Radiation	RadAvg		666	795	1012	1256	1342	1256	1336	1283	932	753	673	628
	RadStd		81	108	101	119	153	154	190	175	103	102	51	82

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (7 neighbors)	+0.52	N/A	-1.13	+0.81	N/A	N/A	-90	-164	+295	+175

Nomenclature: See separate page